



Emotion in Context: Human and AI Perspectives on the Affective Landscape of *Severance*

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Abstract

Emotion detection has long been a focus of interdisciplinary research spanning psychology, linguistics, and discourse analysis. Traditional computational approaches to emotion detection have included both lexicon-based and machine learning-based methods. While lexicon-based approaches offer interpretability and domain independence, they often struggle to capture contextual nuance in complex narratives. Machine learning methods, though effective within their training domains, require substantial annotated data and face challenges with domain adaptation. Recent advances in natural language processing (NLP), particularly the development of large language models (LLMs), offer new possibilities for automating emotion detection with greater contextual awareness. This study investigates how emotions are represented and perceived in the television series *Severance* (Apple TV, 2022) through a multi-layered annotation framework based on Parrott's hierarchical taxonomy of emotions. All nine episodes of season 1 were annotated scene by scene, producing four datasets: human text-only, human multimodal (video+audio+text), and two LLMs. Comparative analyses assessed inter-annotator agreement, modality effects, and temporal and character-based emotion patterns. Results show that multimodal annotations yield a far richer and more nuanced emotional landscape than text alone, highlighting the decisive role of prosody, gesture, and facial expression in conveying affect. LLMs captured general emotional polarity but failed to reproduce fine-grained distinctions. Character profiles further reveal individualized affective strategies of repression, resistance, and awakening. The findings underscore the inherently multimodal nature of emotional meaning in audiovisual narrative.

Keywords Emotion detection · Large language models · Natural language processing · Corpus annotation · TV series

Introduction

Emotions are a central component of human communication, shaping meaning-making processes across cultural contexts. Their study has therefore attracted sustained interest from psychology, linguistics, and discourse analysis. Within computational research, this concern has given rise to affective computing (Picard, 1997), a field that seeks to develop systems capable of recognizing and responding to human emotions. Early approaches drew heavily from psychological models of emotion, particularly dimensional models such as the Valence-Arousal-Dominance (VAD) framework (Russell & Mehrabian, 1977), which provided a foundational approach to quantifying affective states in affective computing.

Within Natural Language Processing (NLP), the task of emotion detection emerged as an extension of sentiment analysis, seeking to identify fine-grained emotions from human language, and thus going beyond the broader polarity distinctions typical of sentiment analysis (Acheampong et al., 2020). Sentiment analysis focuses on the identification and extraction of people's opinions towards other entities and their attributes (Liu, 2015). Emotion detection poses greater challenges due to the complexity of affective meaning (Nandwani & Verma, 2021). Both emotion detection and sentiment analysis, can be approached using lexicon-based or machine-learning based methods, or a combination of both (Moreno-Ortiz et al., 2019). The former represents the simplest approach, which is based on the comparison of the lemmatized tokens in a text with the corresponding lemmas in the lexicon, and giving them an appropriate score (Puschmann & Powell, 2018). While some implementations also include contextual valence shifters and multi-word expressions (e.g., Moreno-Ortiz, 2017), the majority typically follow bag-of-words approaches (Taboada et al., 2011). Although these lexicon-based methods enabled large-scale annotation and quantitative analysis, their effectiveness was constrained by their reliance on word-level associations and their inability to capture contextual, pragmatic, or multimodal dimensions of emotion. In discourse-rich or narrative settings, where emotions may be implied (or even contradicted by sarcasm), such approaches often fall short (Cambria et al., 2017).

Machine-learning models, as explained by Öhman (2021b), are sensitive to context and can be combined with large language models (LLMs) for improved performance. Although they typically perform much better than lexicon-based models (De Clercq et al., 2017; Van Atteveldt et al., 2021), they still perform poorly in domains different to those they have been trained on (Aue & Gamon, 2005; Wang & Liu, 2015).

The development of machine learning and neural network models offered advances such as the detection of complex textual patterns. Techniques such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs) began to account for compositional aspects of language (Socher et al., 2013; Wang et al., 2016), although these models still required large amounts of annotated data and remained limited in their ability to adapt to new domains (Balahur et al., 2012). A major breakthrough came with the introduction of the Transformer architecture (Vaswani et al., 2017). Instead of using recurrent layers, it relies on the self-attention mechanism, which allows the model to handle all parts of a sequence simultaneously. Building on this foundation, BERT (Bidirectional Encoder Representations from Transformers) was

introduced by Devlin et al. (2019), which represented the first deeply bidirectional pre-trained language model. BERT-based architectures rapidly became standard baselines for emotion detection due to their capacity to capture nuanced contextual cues that earlier machine-learning models struggled to identify.

The Transformer architecture is at the core of the current generation of LLMs, collectively referred to as *generative AI*, a term that emerged organically and was originally used in reference to Markov chains models. In a very reduced timeframe, LLMs have transformed all aspects NLP, and introduced new possibilities for all NLP tasks, including emotion detection. Models such as GPT (OpenAI, 2025), LLaMA (Touvron et al., 2023), and Claude (Anthropic, 2025) offer new avenues of research for automating emotion detection with greater contextual awareness, thus attempting to capture contextual nuance, pragmatic inference, and stylistic variation (Khare et al., 2024). Recent research has investigated whether it is possible to automatically annotate texts containing emotions with ChatGPT: Koptyra et al. (2023) reached the conclusion that although possible, ChatGPT's annotations do not reach the same degree of quality of human annotations, while Wake et al. (2023) found a reasonable level of reproducibility in ChatGPT's performance and concluded that the choice of dataset and emotion labels significantly impacts the system's performance. For their part, Latif et al. (2025) achieved improved results through data augmentation by incorporating samples annotated by ChatGPT.

This study compares the performance of two leading generative LLMs (GPT-5 and Claude Sonnet 4.5) in detecting emotions using Parrott's taxonomy (Parrott, 2001). The models were evaluated using a manually annotated gold standard derived from the scripts of *Severance* (Stiller, 2022), a series recognized for its psychological depth and narrative subtlety. Scenes were manually tagged according to the primary, secondary and tertiary emotions expressed by the characters. By evaluating how these models handled the layered, often ambiguous emotional expression of characters, this research offers insight into the current capabilities of generative LLMs for emotion detection and their implications for affective computing and discourse analysis.

This study also aims to provide further insights into the relevance that non-verbal language plays in the expression of human emotions, by comparing two sets human annotations: one where annotators had access exclusively to the scripts (i.e., text-only) and one produced while actually watching the series, thus taking into account verbal and non-verbal expression (i.e., prosody, gestures, facial expressions, posture). The first set of annotations was used to measure the performance of the two LLMs that were evaluated in this study; the second set was used to measure the proportion of verbal versus non-verbal emotion expression.

Theoretical Background

Affective Computing

The computational analysis of human emotions has evolved considerably since Picard (1997) coined the term “affective computing” to describe systems capable of recognizing, interpreting, processing, and inferring human affect. What began as

an ambitious vision of emotionally intelligent machines has developed into a multidisciplinary field drawing on psychology, neuroscience, linguistics, and computer science (Poria et al., 2017). Early affective computing research focused on primarily on rule-based systems and basic pattern recognition. However, the advent of machine learning, particularly deep learning architectures, has enabled more sophisticated approaches to emotion detection. The integration of dimensional models such as VAD with computational methods provided early frameworks for quantifying emotions, while advances in natural language processing and computer vision have expanded the field's capabilities (Zhao et al., 2022).

The capabilities of affective computing largely depend on the modalities employed to infer emotions. In multimodal systems, algorithms go beyond the text and analyze facial expressions and vocal tone to detect affective states (Al-Azani & El-Alfy, 2025). Text-based systems, on the other hand, rely on linguistic cues, which entails significant limitations, such as the ambiguity and subjectivity of language (Mantello et al., 2023; Zahiri & Choi, 2018). Emotions are culturally and contextually shaped, and texts often omit crucial non-verbal cues, such as intonation or facial expressions. Moreover, sarcasm and humor often confound emotion detection algorithms, as these forms of expression depend on pragmatics and inference beyond the literal meaning (Kumar et al., 2022).

By allowing machines to detect and adapt to human emotional states, affective computing has potential applications across different fields, such as healthcare (Cano et al., 2023; Zhou et al., 2023), education (Keerthika et al., 2024), marketing (Pusztahelyi, 2020), social media (Chandrasekaran et al., 2021), or security (Butt et al., 2023), among others.

Emotion Models

Any attempt to manually or automatically classify emotions presupposes the use of an accepted pre-existing schema or taxonomy of emotions. In turn, human emotions present a wide spectrum or continuum of feelings, and arriving at an agreed-upon universal classification is not an easy task, as the selection of emotion categories has itself become a debate. Öhman (2021a) critically examined emotion categorization practices in computational research, and argued that researchers should rethink how emotions are categorized and annotated, as the use of different theoretical frameworks provides distinct emotion inventories with distinct assumptions. Applied projects such as GoEmotions (Demszky et al., 2020), a large-scale dataset for fine-grained emotion classification, demonstrated that the choice of emotion schema had a significant impact on system performance, annotation quality, and the comparability of results across studies.

Dimensional Models

Dimensional models, particularly the Valence-Arousal-Dominance (VAD) framework, have been foundational in affective computing since their introduction (Russell & Mehrabian, 1977). The VAD model represents emotions as points in a three-dimensional space: valence (pleasure-displeasure), arousal (activation-deactivation),

and dominance (control-lack of control). This approach has proven particularly valuable for computational applications because it provides a continuous representation of emotional states, allowing for the modeling of subtle affective variations and the mapping of discrete emotion categories onto a dimensional space. VAD-based models remain widely used in affective computing for tasks such as emotion recognition from speech, facial expressions, and physiological signs, and many emotion lexicons (e.g., ANEW, EmoLex) include VAD ratings for words (Mohammad & Turney, 2013). The dimensional approach has also been integrated with modern transformer-based models, enabling systems to predict both discrete emotion categories and continuous affective dimensions simultaneously.

Categorical Models

Categorical models define emotions discretely, thus classifying them into distinct categories according to the particular model being used. Ekman's model, one of the most salient in the literature, distinguishes six basic emotions: happiness, surprise, anger, disgust, sadness, and fear (Ekman, 1992).

Plutchik's wheel of emotions (Plutchik, 2001) proposes eight primary emotions (joy, trust, fear, surprise, sadness, disgust, anger, and anticipation) that serve as the foundation for all others. These emotions are arranged as four pairs of opposites, based on the physiological reaction each emotion creates in animals (joy-sadness, trust-disgust, fear-anger, surprise-anticipation). The wheel is also dimensional, as each primary emotion can vary in intensity and can be blended to express nuanced affective states.

Parrott's classification (2001), in turn, presents a tree-structured list of emotions based on six primary ones: love, joy, surprise, anger, sadness, and fear, as shown on Table 1. Parrott's model is particularly useful for linguistic and discourse-based research, as it provides a wide array of emotion labels for the annotation of emotional nuance in text. The availability of this large number of labels facilitates annotation work but poses certain difficulties when it comes to calculating agreement. However, its hierarchical design, which maps lower-order emotions to higher-order ones, makes it possible to calculate agreement at different levels. Table 1 contains the full set of emotion labels in Parrott's model, including the mappings.

The Human-Machine Interaction Network on Emotion (HUMAINE) project proposed the emotion annotation and representation language (EARL), a framework designed to provide a standardized format for describing emotions across different modalities. It considers 49 emotions grouped into 10 main ones: reactive, quite positive, positive thoughts, positive and lively, caring, agitation, negative and passive, negative thoughts, negative and not in control, and negative and forceful (Douglas-Cowie et al., 2011).

Recent developments in psychology have introduced new ways, both conceptual and methodological, to model the semantic space of emotions. In this line, Cowen and Keltner (2017) developed an emotion categorization consisting of 27 emotion clusters, with fuzzy rather than firm boundaries, by studying people's emotion responses to a wide variety of stimuli. This approach acknowledges the gradient nature of emo-

Table 1 Parrott's emotion classification

Primary emotion	Secondary emotion	Tertiary emotion
Love	Affection	Adoration/fondness/liking/attraction/caring/tenderness/compassion/sentimentality
	Lust	Desire/passion/infatuation
	Longing	Longing
Joy	Cheerfulness	Amusement/bliss/gaiety/glee/jolliness/joviality/joy/delight/enjoyment/gladness/happiness/jubilation/elation/satisfaction/ecstasy/euphoria
	Zest	Enthusiasm/zeal/excitement/thrill/exhilaration
	Contentment	Pleasure
	Pride	Triumph
	Optimism	Eagerness/hope
	Enthrallment	Enthrallment/rapture
	Relief	Relief
Surprise	Surprise	Amazement/astonishment
Anger	Irritability	Aggravation/agitation/annoyance/grouchy/grumpy/crosspatch
	Exasperation	Frustration
	Rage	Anger/outrage/fury/wrath/hostility/ferocity/bitterness/hatred/scorn/spite/vengefulness/dislike/resentment
	Disgust	Revulsion/contempt/loathing
	Envy	Jealousy
Sadness	Torment	Torment
	Suffering	Agony/anguish/hurt
	Sadness	Depression/despair/gloom/glumness/unhappiness/grief/sorrow/woe/misery/melancholy
	Disappointment	Dismay/displeasure
	Shame	Guilt/regret/remorse
	Neglect	Alienation/defeatism/dejection/embarrassment/homesickness/humiliation/insecurity/insult/isolation/loneliness/rejection
	Sympathy	Pity/mono no aware/sympathy
Fear	Horror	Alarm/shock/fear/fright/horror/terror/panic/hysteria/mortification
	Nervousness	Anxiety/suspense/uneasiness/apprehension (fear)/worry/distress/dread

tional experience while maintaining categorical distinctions for practical classification purposes.

Hybrid Models

An approach that bridges categorical and dimensional frameworks is the Hourglass of Emotions (Cambria et al., 2012). This model represents affective states through both discrete emotion labels and four independent but concomitant affective dimensions: pleasantness, attention, sensitivity, and aptitude. Each dimension contains six levels of activation. It was specifically designed to address limitations of purely categorical or dimensional approaches in applications such as sentiment analysis and human-computer interaction, and it forms the theoretical foundation for SenticNet, a large-scale semantic resource that combines commonsense knowledge with affective information for concept-level sentiment analysis (Cambria et al., 2018).

Practical Considerations in Model Selection

As in any classification task, reducing the number of categories generally simplifies the process. Consequently, emotion detection tasks, such as those organized by SemEval (Chatterjee et al., 2019; Kumar et al., 2024), tend to adopt simpler emotion models with fewer categories. Although the most widely used frameworks in computational emotion detection are Ekman's and Plutchik's, debate persists within the research community regarding which categorical model best captures human emotions (Öhman, 2021a). This has led researchers to frequently adapt existing taxonomies, combine elements from different frameworks, or develop custom emotion categories tailored to their particular datasets and research objectives: for example, Demszky et al. (2020) used emotions that were detected as salient by Cowen and Keltner (2017) while making sure that their set included Ekman's emotion categories, but also allowed annotators to suggest emotions that were not among the given options.

This study employs Parrott's schema because of the hierarchical arrangement it proposes, which allows the evaluation of the classification results at the primary level even when the LLM produces secondary or tertiary emotion categories during automatic classification. In addition, the model's extensive vocabulary of emotion labels is well-suited to capturing the emotional expressions characteristic of dramatic dialogue in narrative contexts, and it provides interpretable labels that facilitate both human annotation and the evaluation of LLM performance in text-based emotion detection tasks.

In order to obtain agreement metrics using this this multi-level schema, a custom script was written that re-created the underlying tree structure and was able to calculate agreement at any level for any emotion label contained in the schema.

Large Language Models

Current LLMs are sophisticated statistical pattern recognizers and generators, trained on vast amounts of textual data to predict and generate human-like language patterns (Minaee et al., 2024). As mentioned in the introduction section, their origins extend back to transformer architectures such as BERT (Devlin et al., 2019), whose pre-training objectives and contextual embeddings form the foundation on which many contemporary LLMs are built. They can perform tasks such as text generation, image description, or question answering, among others (Phogat et al., 2025). Their capacity to generate coherent, contextually relevant responses has made them central to modern NLP and conversational systems.

As explained by Zhao et al. (2023), the concept of language models has evolved with the advance of artificial intelligence over the decades. While early LLMs mainly aimed to model and generate text data, latest LLMs focus on complex task solving. The rapid progress of such models has had an impact on the growing number of organizations that now produce systems capable of generating coherent and contextually appropriate outputs from natural-language prompts. This expansion stems from several factors, as open-source initiatives have made some datasets more easily accessible to academic researchers, such as *The Pile*, a corpus targeted at training

large-scale language models (Gao et al., 2021). Moreover, the understanding of transformer architectures has matured and demand has grown, thus encouraging continuous innovation (Zhao et al., 2023). Consequently, the proliferation of LLMs has led to a rich ecosystem in which many competing models coexist, such as DeepSeek, Qwen, ChatGPT, or Claude Sonnet. This study, however, will concentrate on the last two, which represent distinct yet complementary visions of what an LLM can be and do.

Limitations of Current Large Language Models

Despite their impressive capabilities, current large language models exhibit several critical limitations that constrain their reliability and trustworthiness in real-world applications. Hallucinations represent one of the most significant challenges, where LLMs generate factually incorrect or entirely fabricated information with apparent confidence. Hallucination is not merely a technical flaw to be engineered away, but rather an inherent limitation of the statistical learning paradigm underlying these models (Xu et al., 2025). Measuring the impact of this issue, however, is extremely complex, as it varies enormously depending on the task and domain. For example, in legal contexts, LLMs have been shown to hallucinate at least 58% of the time when answering case law questions (Dahl et al., 2024). Similarly, medical applications reveal systematic vulnerabilities to adversarial attacks that trigger hallucinations in clinical decision support scenarios (Omar et al., 2025). In other less specialized tasks, modern LLMs display significantly lower rates of hallucinations.

A second, well-known issue is that of sycophancy, which refers to LLMs' tendency to prioritize user agreement over accuracy, generating responses that confirm users' beliefs rather than providing objective information, which creates a tension between alignment with user expectations and factual accuracy, particularly when those expectations include false premises. Studies evaluating GPT-4o, Claude Sonnet, and Gemini found overall sycophancy rates of 58.19%, with this behavior particularly pronounced when users present claims with high confidence or cite perceived authorities (Fanous et al., 2025; Rrv et al., 2024).

Thirdly, social biases persist systematically across LLMs, reflecting and often amplifying biases present in training data (Gallegos et al., 2024). Recent research demonstrates that even models designed to pass explicit bias tests continue to harbor implicit biases across race, gender, religion, and other social categories (Bai et al., 2025). These biases manifest in discriminatory decisions, with evaluation studies showing significant differential treatment based on demographic characteristics in critical contexts such as resume evaluation (An et al., 2025).

In terms of reasoning, limitations are apparent in moral and logical decision-making contexts. LLMs have been shown to exhibit amplified cognitive biases compared to humans, including stronger omission bias (preferring inaction over action) and a tendency to flip decisions based on question framing rather than logical consistency (Cheung et al., 2025). These reasoning failures are particularly concerning as they can lead to systematically flawed advice in contexts requiring careful deliberation.

Finally, context sensitivity further constrains LLM performance, with evidence that simple changes to system instructions (i.e., the prompt), such as requesting con-

cise responses, can degrade factual reliability by up to 20% (Le Jeune & Berenstein, 2025). This brittleness reflects LLMs' limited understanding of pragmatic context and their tendency to prioritize stylistic constraints over accuracy.

These limitations have substantial implications for the use of LLMs in critical domains and expose the need for careful validation and continued research. In the context of the current study, we attempt to provide an objective measurement of the performance of some of the best LLMs available today in relation to emotion detection in telecinematic discourse.

LLMs Used in this Study: ChatGPT and Claude

The evaluation of large language model performance relies on a diverse ecosystem of standardized benchmarks designed to assess distinct capabilities across multiple domains. Among the most influential are MMLU (Massive Multitask Language Understanding), which evaluates breadth of knowledge across 57 subjects ranging from STEM fields to humanities (Hendrycks et al., 2021). Reasoning benchmarks include HellaSwag, which tests commonsense reasoning through plausible sentence completions (Zellers et al., 2019), and BBH (Big-Bench Hard), a subset of particularly challenging tasks from the larger Big-Bench collection (Suzgun et al., 2023). Conversational abilities are evaluated through MT-Bench, which assesses multi-turn dialogue quality (Zheng et al., 2023), and the Chatbot Arena, a crowdsourced platform where models compete via human preference comparisons (Chiang et al., 2024). To our knowledge no specific benchmark exists for emotion detection. The two LLMs we will be evaluating for emotion detection, GPT-5 and Claude Sonnet 4.5, are amongst the top-ranked LLMs in these and other benchmarks.

ChatGPT¹, developed by OpenAI (2025), is perhaps the most widely known chatbot. The underlying language models have been pre-trained on massive amounts of text data and then fine-tuned to behave as a useful, human-like, assistant. The model encodes prompts as word embeddings, extends context via transformer layers, and generates responses token by token. GPT-5, the latest milestone in OpenAI's line of generative-pretrained-transformer models, is described by its developers as their smartest and most useful model yet. Unlike earlier models that often excelled only within narrow domains and left it to the user to select a specific model for each task, GPT-5 is intended to perform at a high level across multiple domains and it automatically performs optimal sub-model selection. Despite its achievements, GPT-5 is still subjected to limitations, as it can generate plausible but incorrect information (sometimes known as "hallucinations").

Claude is a series of LLMs developed by Anthropic (2025). They have been pre-trained to make predictions in large amounts of text, and have been fine-tuned using constitutional AI and reinforcement learning from human feedback (Henshall, 2023). Sonnet 4.5 is the latest iteration in the "Sonnet" line of Claude models and it is designed for autonomy, extended context handling, and integration into complex workflows. While GPT-5 aims for broad general performance across domains, Sonnet

¹ We will use "ChatGPT" when referring to the application or its interface, and "GPT-5", i.e., the model name followed by the version number, when referring to the language model itself.

4.5 is tailored for extended, practical problem-solving scenarios that require persistence, sustained attention, and numerous intermediate steps. This makes it interesting for research and development contexts that require consistency and long-term state tracking, but it might also distance it from what casual users are looking for.

When viewed together, GPT-5 and Claude Sonnet 4.5 exemplify two complementary approaches in the current evolution of LLMs (Jiang et al., 2025). GPT-5 represents the generalist vision: it is a broad language model intended for everyday use across domains. Claude Sonnet 4.5, on the other hand, represents the specialist approach: a focused tool-using system designed for deep, sustained engagement.

Research Design

Objectives and Methods

This work investigates how emotion is represented and perceived in narrative television discourse, and how analytical modality (text-based, multimodal, or computational) shapes emotional interpretation. This main goal translates into three specific objectives:

1. To compare text-only versus multimodal human emotion annotation, thus assessing how access to visual and auditory information might alter emotional perception.
2. To evaluate the performance of LLMs in identifying emotions from dialogue scripts, benchmarking them against human annotations.
3. To map the emotional landscape of *Severance*, analyzing the distribution and evolution of emotions across episodes and characters following Parrott's taxonomy.

Corpus Description

The corpus consists of the complete first season of *Severance* (Stiller, 2022), encompassing nine episodes, 335 scenes, and 49 characters. The dialogue comprises 25,612 words distributed across 3062 individual dialogue entries. The textual component of the corpus was derived from subtitle (SRT) files, which were obtained from Open-Subtitles. We used regular expressions to remove all data except the text itself, character information, and timecodes, resulting in one text file per episode. The files were manually analyzed to identify scenes. Each episode was then converted into a structured JSON file that encoded the episode and scene identifiers, the participating characters in each scene, and an initially empty list of emotions to be filled during annotation. This uniform JSON structure served as the common input for both human annotators and the language models. Timecodes were preserved for future reference, although they were irrelevant in the present study.

Annotation Scheme and Gold Standards

Emotion categories followed Parrott's (2001) hierarchical model, comprising three levels, as shown on Table 1. Each scene could contain one or more emotions per character, depending on the expressive content of the dialogue and, in the multimodal case, accompanying visual or prosodic cues. Annotators were also asked to choose the most specific emotion possible or select a higher-order one if unsure.

Two parallel gold standards were developed to capture different perspectives of emotional interpretation: (i) a text-only gold standard (based exclusively on the dialogue scripts), and (ii) a multimodal gold standard (based on the audiovisual episodes). Both were produced through a rigorous three-stage process designed to guarantee reliability and validity:

1. Independent annotation: two trained annotators independently labeled the emotions expressed by each character in every scene, using a JSON-aware text editor.
2. Inter-annotator agreement: agreement between annotators was calculated using Cohen's Kappa (κ) coefficient (1960) for each emotion level and modality.
3. Adjudication: disagreements were resolved through a union strategy, by which any emotion identified by either annotator was retained in the final gold standard.

This conservative approach guaranteed maximal recall and preserved the richness of emotional interpretation, especially for subtle or overlapping states. As a result, we obtained two high-quality, scene-level annotated corpora (one text-based and one multimodal) serving as gold standards for model evaluation and comparative analysis.

LLM Annotation and Evaluation

The same JSON files were provided to two LLMs—LLM 1 GPT-5 (OpenAI); LLM 2: Claude Sonnet 4.5 (Anthropic)—using the same prompt:

"You are a corpus annotator. I will give you a number of JSON files with the scripts of the series *Severance*. Each file corresponds to an episode. Your aim is to decide which emotions are expressed in the text by the characters in each scene. You will find a data structure like this "emotions": {"Mark": [], Helly": []}. This is placed before each scene's dialogue. I want you to fill the empty list of emotions for each character in that scene, as expressed by their words. You must use Parrott's emotion framework, which classifies emotions as a hierarchical, 3-level tree. Before you start annotating, I will give you the precise labels you must use as a Python function that I have created to build the tree structure and allow me to map lower to higher-level emotions. Make sure you understand these instructions before proceeding."

GPT-5's response (after uploading the tree-building Python function):

"Perfect—I've read and understood your `construct_emotion_tree()` function and the exact labels it defines. I will:

- Use only the labels present in that tree (either top-level or leaf-level labels as appropriate).

- Fill each scene's "emotions": { ... } lists with one or more of those labels.
- Leave a character's list empty if no clear emotion is evident.
- Return the same JSON structure with the emotion lists populated.

Go ahead and upload (or paste) one episode JSON file and I'll annotate it right away."

Claude's response:

"Perfect! I now have the complete Parrott emotion framework with all the hierarchical levels. I can see the structure clearly:

- Level 1 (Primary): love, joy, surprise, anger, sadness, fear.
- Level 2 (Secondary): affection, lust, longing, cheerfulness, zest, contentment, pride, optimism, enthrallment, relief, irritability, exasperation, rage, disgust, envy, torment, suffering, disappointment, shame, neglect, sympathy, horror, nervousness.
- Level 3: finer-grained emotions.

I'm ready to annotate! Please share the JSON files with the Severance episode scripts, and I'll:

1. Analyze each scene's dialogue.
2. Identify which emotions each character expresses.
3. Fill in the emotion lists with the appropriate labels from this framework.
4. Use the most specific/granular level that accurately captures the emotion.

Go ahead and send the first JSON file when you're ready!"

This prompt instructs LLMs to read the dialogue content and character metadata and to populate the empty emotion lists with the emotions expressed by each character in each scene. Thus, the models operated in a text-only setting, without visual input.

Their outputs were directly compared to the human text-only gold standard using standard binary metrics (precision, recall, and F1-score computed per emotion and per hierarchical level). Confusion matrices and macro-averaged scores were used to quantify performance and to assess model sensitivity to specific emotional categories.

Subsequently, the following analyses were carried out:

1. Human modality comparison (text vs. multimodal): overall emotion distributions, prevalence shifts, and correlation between modalities.
2. LLM performance evaluation: precision, recall, F1, and confusion profiles across hierarchical levels.
3. Character-level emotion patterns: how emotional profiles vary by character and modality.
4. Temporal evolution: episode-by-episode trajectories of key emotions and divergence between modalities over time.

These analyses provided a multi-layered picture of how emotions operated in *Severance*, from script to screen to algorithmic interpretation.

Data Processing and Analytical Pipeline

As mentioned above, the annotation workflow was built on a digital corpus derived from the episode subtitle files. Each subtitle (SRT) file was first segmented into discrete scenes, defined by changes in location, time, or conversational focus. Scene segmentation was manually checked against the episode timeline to ensure narrative coherence. All dialogue lines were then aligned with their corresponding scene and speaker information. The data were exported into JSON format, one file per episode, with the core structure illustrated in Fig. 1:

This standardized schema facilitated interoperability across annotation environments and ensured that both human annotators and language models worked on identical inputs. The use of JSON allowed structured storage of scene-level information (episode, characters, dialogues, and assigned emotions) while remaining lightweight enough for direct model ingestion.

Annotation Output and Post-processing

Each annotated JSON file was automatically validated to guarantee consistent label formatting and completeness. For each hierarchical level in Parrott's taxonomy (primary, secondary, and tertiary), the following data were extracted from the annotated JSON data and tabulated into several CSV files to more easily address the specific objectives:

1. Character-level emotion counts and percentages per scene and episode.
2. Modality indicators (text vs. multimodal; LLM1 vs. LLM2).
3. Episode-wise emotion frequencies.
4. Aggregated distributions for global, character, and temporal analyses.

A Python pipeline using pandas handled this transformation, computing summary statistics and generating derived datasets for downstream analysis (e.g., confusion matrices, performance metrics, temporal trends).

```
{
  "episode": "S01E01",
  "scene_number": 12,
  "characters": ["Mark", "Helly"],
  "dialogues": [
    {"timecode": "00:05:14", "character": "Mark", "text": "Who are you?"},
    {"timecode": "00:05:22", "character": "Helly", "text": "I don't understand."}
  ],
  "emotions": {}
}
```

Fig. 1 Core structure of the data

Table 2 IAA results for the text-only gold standard

Level	Total comparisons	Agreements	Disagreements	Cohen's Kappa	Interpretation
Primary	2010	1954	56	0.9131	Almost perfect
Secondary	7370	7242	128	0.8284	Almost perfect
Tertiary	19,430	19,179	251	0.6313	Substantial

Table 3 IAA results for the multimodal gold standard

Level	Total comparisons	Agreements	Disagreements	Cohen's Kappa	Interpretation
Primary	2010	1939	71	0.9128	Almost perfect
Secondary	8375	8152	223	0.8250	Almost perfect
Tertiary	30,820	30,343	477	0.0596	Moderate

Data Analysis

Annotation Process and Gold Standards

Inter-annotator agreement was assessed using Cohen's Kappa (κ), calculated independently for each level of Parrott's hierarchical emotion taxonomy. The analysis measured the extent to which two human annotators independently agreed on the presence or absence of each emotion across all scene-character combinations.

To prepare the data, tertiary emotion labels were mapped upward to their corresponding categories at the upper hierarchical levels. For each scene-character pair, binary indicators were created for every emotion category (e.g., at the secondary level: Does this character express fear? $\rightarrow$ Yes = 1, No = 0). Cohen's Kappa was then computed across all such binary decisions, providing an aggregate measure of agreement beyond chance. Tables 2 and 3 display the results for the text-only gold standard and the multimodal gold standard, respectively².

Across all levels and modalities, inter-annotator agreement is very high, thus confirming that the annotation schema and guidelines were applied consistently by both raters. The almost perfect agreement at the primary and secondary levels indicates clear, shared understanding of broad emotional categories (e.g., love, joy, surprise, sadness, anger, fear) and their secondary derivatives. However, finer-grained distinctions (e.g., anxiety vs. insecurity, frustration vs. irritability) introduce more subjectivity and ambiguity, reducing reliability slightly but still well above chance.

Together, these results confirm that the annotation framework is both reliable and modality-robust: human interpretation remains consistent whether emotions are inferred from text alone or from the full audiovisual context. Contrary to what might be expected, access to visual and auditory cues did not significantly increase reliability; κ values are nearly identical or slightly lower for the multimodal condition at every level. The reason for this may be that the multimodal task required simultaneous attention to multiple channels, potentially increasing cognitive load and small divergences in decision boundaries.

² Interpretive scale: $\kappa < 0.00$ = Poor, 0.00 – 0.20 = Slight, 0.21 – 0.40 = Fair, 0.41 – 0.60 = Moderate, 0.61 – 0.80 = Substantial, 0.81 – 1.00 = Almost perfect.

Performance of LLMs

To assess how well language models can recognize emotions from dialogue alone, two large language models (GPT-5 and Claude Sonnet 4.5) were asked to identify the emotions expressed by the characters in *Severance*, based only on the episode scripts. Their results were then compared with those of human annotators who had carried out the same task using only the text of the scripts (the text-only gold standard).

Because emotions were annotated using Parrott's hierarchical model, the models' performance was evaluated separately at each level of granularity. Table 4 summarizes the results using different metrics to compare performance.

Even though these numbers look small at first glance, they are quite revealing. Cohen's Kappa (κ) tells us how closely a model's decisions match those of human annotators after correcting for chance. A value near 1 means near-perfect agreement; 0 means agreement no better than random guessing. In our case, κ values between 0.10 and 0.21 show that the models captured some emotional structure, but their understanding remained limited compared to humans.

F1-scores reflect how often the model correctly identified the right emotion when it appeared. Claude's F_1 -scores are consistently higher than GPT-5's, meaning it was more balanced in correctly identifying both frequent and rare emotions. Accuracy is high (around 90%), but this can be misleading: since many lines contain no emotion, simply predicting "none" can inflate this value. The more meaningful measures are κ and F1, which show how well the model performs when emotions actually occur.

Therefore, Claude Sonnet 4.5 consistently achieves slightly higher agreement with human annotations, especially at the secondary level, where its advantage is most visible. This suggests that this LLM may be better at handling moderately complex emotional categories such as affection, irritation, or worry. While the numerical improvements may appear modest (e.g., κ rising from 0.13 to 0.16), they are consistent across all levels and suggest Claude produces more human-like judgments overall.

Both models perform best for broad, easily recognizable categories like anger or fear, and struggle with subtle distinctions such as amusement versus contentment. This mirrors what is often observed in human annotation: the finer the categories, the greater the ambiguity.

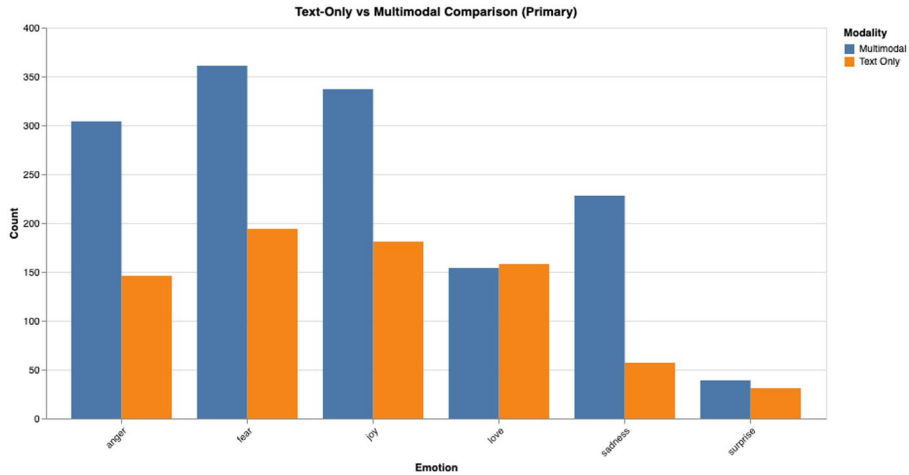
Finally, both models tend to recognize some emotions better than others. Anger, love, and fear were the most accurately detected, while surprise, joy, and amusement

Table 4 Performance comparison of GPT-5 and Claude sonnet 4.5 versus human gold standard (values in bold represent best results)

Level	Model	Agreement (κ)	F_1 (weighted)	F_1 (per emotion)	Accuracy
Primary	GPT-5	0.13	0.33	0.21	0.67
	Claude Sonnet 4.5	0.16	0.35	0.23	0.70
Secondary	GPT-5	0.14	0.20	0.16	0.85
	Claude Sonnet 4.5	0.21	0.26	0.15	0.90
Tertiary	GPT-5	0.10	0.13	0.08	0.94
	Claude Sonnet 4.5	0.11	0.13	0.14	0.96

Table 5 Total annotation by parrott's 3-level hierarchy of emotions

Level	Text-only total	Multimodal total	Num. emotions
Primary	767	1423	6
Secondary	663	1283	25
Tertiary	529	954	95

**Fig. 2** Comparison of text-only versus multimodal by number of annotations—primary emotions

were frequently missed—likely because these depend heavily on tone of voice or visual cues rather than the wording alone.

Modality Comparison: Text-Only Versus Multimodal Human Annotation

To assess the extent to which emotional meaning depends on language, as opposed to multimodal expression, we compared human annotations carried out from the scripts alone (text-only) with those produced while watching the episodes (multimodal). The text-only annotations serve as the baseline, allowing us to measure how visual and auditory cues alter emotional interpretation.

Overall, the multimodal mode produced substantially more emotion annotations at all hierarchical levels of Parrott's model (Table 5). The total number of identified emotions increased from 767 to 1423 at the primary level, from 663 to 1283 at the secondary level, and from 529 to 954 at the tertiary level. This indicates that non-verbal cues such as prosody, facial expression, gesture, and posture reveal additional emotional nuances that remain implicit or ambiguous in the written dialogue.

In terms of emotions, more annotations were generated across the board for the multimodal corpus. There were some differences, however. Figures 2 and 3 show the difference in number of annotations for the primary and secondary emotions, respectively.

These counts clearly indicate that humans do encode much of their emotional expressions through means other than language, but apart from this rather obvious observation, there are some interesting finer details. Fear rises from 194 to 361

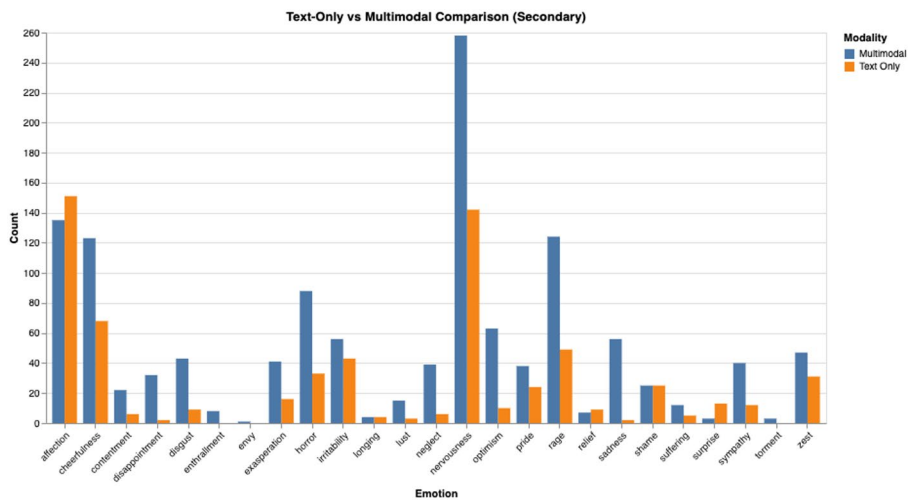


Fig. 3 Comparison of text-only versus multimodal by number of annotations—secondary emotions

annotations, and anger more than doubles from 146 to 304. Joy also increases considerably, whereas love remains roughly stable, indicating that affection is largely captured through dialogue rather than non-verbal delivery. Sadness exhibits the sharpest relative increase (57–228), suggesting that this emotion is often conveyed through facial expression, tone, or silence rather than explicit wording. Finally, surprise rises only slightly (31–39), implying that its textual markers (e.g., “what?”, “really?”) are already strong cues.

At the secondary level, both emotional diversity and the magnitude of change increase. Here, 25 categories such as nervousness, rage, affection, and sadness show significant shifts. The largest absolute increase occurs for nervousness (142–258), followed by rage (49–124) and horror (33–88). These are intense, reactive emotions that depend strongly on delivery—tone of voice, physical agitation, or facial tension. Sadness expands dramatically (2–56), echoing the pattern seen at the primary level. Optimism (10–63) and cheerfulness (68–123) also rise, indicating that positive affect is more easily detected through performative cues (smiles, relaxed tone) than text. Conversely, affection slightly decreases (151–135), showing that linguistic expressions of care (“thank you,” “you’re kind”) may not always translate into visible emotion on screen. Some nuanced categories such as surprise and sympathy change direction, reflecting contextual shifts in how these emotions are performed.

The linguistic signal in *Severance*, therefore, provides a reliable scaffold for identifying broad emotions, but the multimodal signal (e.g., prosody, expressions, gestures) unlocks the fine-grained, socially embedded dimensions of emotion that are central to the series’ tone and narrative style. Where the script outlines emotional intent, performance actualizes it: fear becomes anxiety, anger becomes frustration, love becomes caring, and so on.

Thus, the characters in *Severance* communicate much of their emotional meaning through performance rather than language. While the script establishes the general emotional trajectory, the affective nuances emerge through tone of voice, facial

expression, and gesture. This multimodal enrichment is especially visible in subtle or ambivalent emotions, such as anxiety, frustration, or tenderness, that depend on delivery rather than wording. The comparatively restrained dialogue style of *Severance* aligns with its thematic focus on repression, surveillance, and controlled affect, making the non-verbal channel essential for decoding the characters' true emotional states. In this sense, the multimodal analysis not only quantifies the added expressivity of audiovisual cues but also highlights the performative economy of the series: emotional intensity is conveyed less through words and more through embodied cues, silence, and micro-expressions that reveal what the text conceals.

Results

Evolution of Emotions Over Time

This section examines how the emotional landscape of *Severance* evolves throughout its first season, tracing changes in the prevalence of different emotions across episodes. The goal is to identify major affective trajectories (e.g., increasing tension, emotional shifts, or recurring patterns) that structure the narrative progression of the series.

Since earlier analyses established that multimodal annotations provide a richer and more comprehensive account of emotional expression, this longitudinal study focuses exclusively on the multimodal data. The inclusion of visual and auditory cues captures subtle affective nuances (e.g., tone, hesitation, facial micro-expressions) that are often absent or ambiguous in the dialogue alone. By concentrating on these multimodal annotations, we obtain a more valid picture of how emotions are performed and perceived in the audiovisual medium, thus revealing the dynamic affective architecture that underlies *Severance*'s distinctive tone and dramatic pacing. Figures 4 and 5 visually represent how emotions evolve over time. Figure 4 shows the evolution of

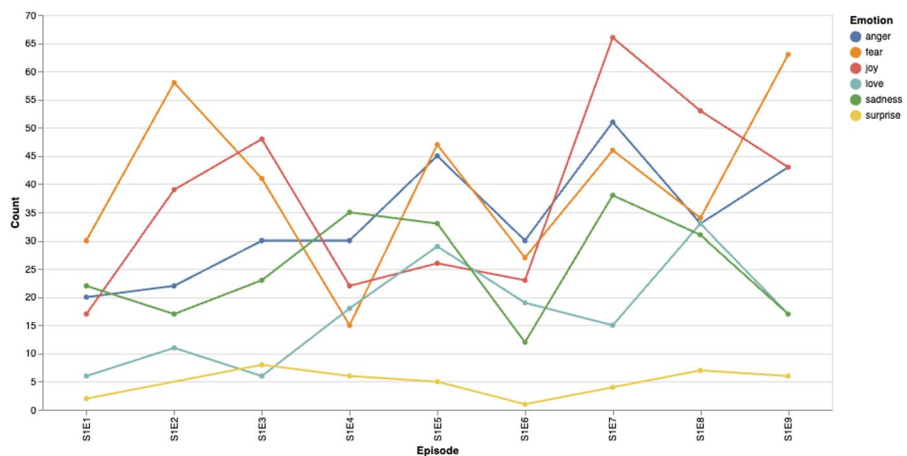


Fig. 4 Evolution of primary emotions

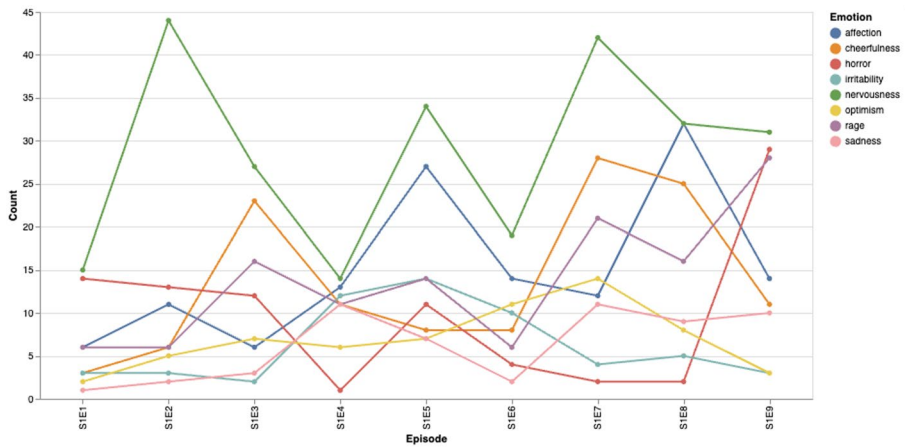


Fig. 5 Evolution of secondary emotions (top 8)

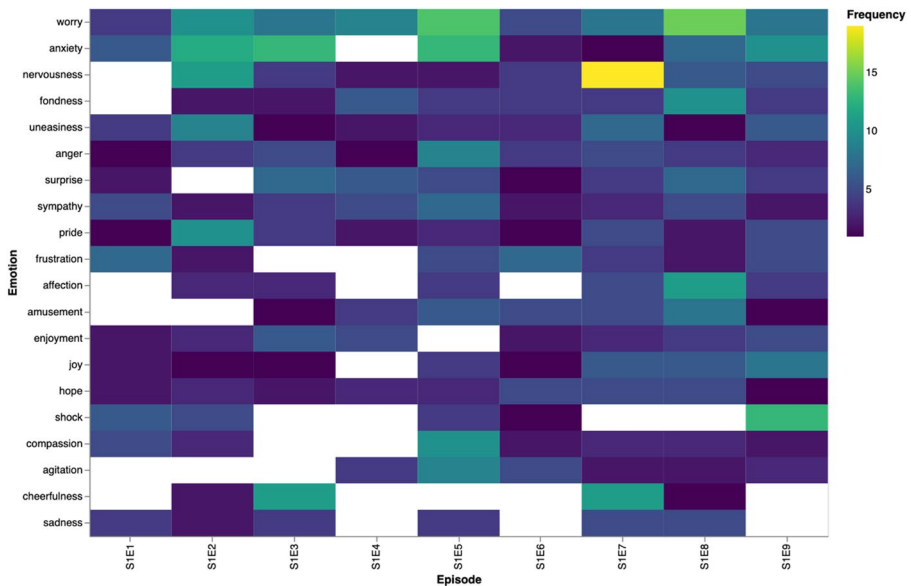


Fig. 6 Heatmap of top 20 emotions per episode

emotions at the primary level, Fig. 5 displays secondary emotions, and Fig. 6 visualizes the 20 most frequent emotions per episode as a heatmap.

Across the season, the multimodal annotations reveal systematic emotional build-up: fear and anger intensify toward the mid-to-late episodes, while sadness accumulates near the finale, consistent with escalating risk and loss. At the secondary level, nervousness, rage, and horror show the clearest growth, indicating progressive erosion of psychological safety. Fine-grained tertiary states (especially anxiety, frustration, uneasiness, and fondness) trace micro-arcs that punctuate the narrative

Table 6 Top three emotions in the main characters

Character	Primary emotions	Secondary emotions	Tertiary emotions
Helly	Fear, anger, sadness	Nervousness, rage, frustration	Anxiety, despair, panic
Mark	Sadness, fear, love	Nervousness, affection, optimism	Anxiety, grief, compassion
Milchick	Joy, anger, fear	Cheerfulness, rage, disgust	Excitement, frustration, bitterness
Irving	Love, fear, sadness	Affection, nervousness, cheerfulness	Fondness, admiration, uneasiness
Cobel	Anger, fear disgust	Rage, nervousness, cheerfulness	Bitterness, contempt, anxiety

with alternating tension and fragile intimacy. Episodic spikes in panic and outrage align with key plot turns, and percentage-normalized trajectories confirm that these changes are not merely artifacts of episode length. Overall, the multimodal temporal profiles portray a slow-burn emotional trajectory dominated by rising threat and fraying control, softened intermittently by brief surges of affiliative affect.

Taken together, the episode-by-episode trajectories reveal a progressive intensification of emotional tension across the season, accompanied by an increasing diversification of affective states. The dominant trend is a shift from initial emotional restraint, characteristic of the series' controlled and depersonalized setting, to growing expressions of fear, anger, and anxiety, reflecting both the characters' psychological unravelling and the gradual exposure of underlying conflicts. At the same time, isolated peaks of fondness, optimism, and joy punctuate the narrative, offering brief but significant counterpoints to the prevailing tone of unease. This oscillation between containment and eruption constitutes a defining feature of *Severance's* emotional architecture, where affect is not evenly distributed but strategically modulated to mirror the series' thematic concerns with repression, identity, and awakening. The multimodal analysis thus captures the show's emotional crescendo, culminating in a finale where affective intensity and narrative revelation converge.

Character-Based Emotional Profiles

This section examines the emotional makeup of the principal characters in *Severance* according to the multimodal annotations. Whereas earlier analyses focused on global distributions and temporal trends, here the aim is to capture the individual affective identities that emerge through patterns of expression across the series. The multimodal data reveal how each character's emotions are conveyed within Lumon's constrained environment. By aggregating the annotations at primary, secondary, and tertiary levels, we obtain distinctive emotional signatures for each protagonist, reflecting both their psychological roles and their evolution over time.

Each of the main characters in *Severance* reveals a distinctive emotional trace that aligns with their psychological arcs and roles within Lumon's hierarchy. Table 6 summarizes the emotional profiles of the five main characters of the series, including their global emotional profile and how this evolves as the season unfolds.

Helly's emotional profile is dominated by fear, anger, and sadness. These data reflect her function as the emotional catalyst of the series: her panic and defiance are not only reactions to confinement, but also a sustained emotional stance against the

dehumanization of the severance procedure. The multimodal annotations, especially the strong presence of high-arousal negative affects, suggest that her resistance is grounded in embodied distress (tone, gestures, facial tension) and verbal rebellion.

Mark presents a more internally conflicted emotional arc. His primary emotions are sadness, fear, and love, which signals ongoing anxiety and cognitive dissonance balanced by affection and optimism. At tertiary level, his anxiety and compassion emerge as markers of a restrained yet emotionally responsive figure. Thus, the multimodal evidence portrays Mark as a man caught between emotional numbness and empathy.

Milchick's emotions show ambivalence, as cheerfulness and optimism coexist with anger and fear. The contrast underscores the performative dimension of his affective behavior—his forced enthusiasm conceals coercive tension. The multimodal data suggest deliberate emotional modulation: wide smiles and controlled body language attempting to hide authority and frustration.

Irving presents a complex profile marked by love, fear, and sadness, complemented by tertiary expressions of attraction, agitation, and fondness. This combination reflects his dual nature: an obedient worker and an individual capable of genuine emotional connection. The coexistence of order and longing is statistically clear, as positive affect is frequent, but always tempered by tension-related states.

Finally, Cobel's emotional distribution is skewed toward rage, nervousness, and disgust, with tertiary clusters of bitterness, contempt, and anxiety. Her multimodal portrayal is that of controlled volatility: minimal verbal emotion but intense nonverbal signaling (glances, posture, tone). These results capture a psychological portrait of someone who maintains authority through suppression and fear.

Overall, the emotional profiles of the main characters reveal a consistent emotional system in which individual emotional patterns intersect with institutional control. The analysis shows that Lumon's world is not emotionless but strategically contained, as each character embodies a different response to emotional suppression. Helly shows her resistance through visible distress, Mark's muted emotions reflect his divided identity, Milchick performs forced enthusiasm as a mechanism of control, Irving turns obedience into melancholy, and Cobel hides moral conflict behind self-control. Together, these examples show how the TV series connects emotional behavior with systems of control: expression and repression, human fragility and corporate order. This supports the view that *Severance* is not only a dystopian narrative about divided identities, but also about how emotions are regulated as a form of power.

Limitations

This study provides valuable insights into emotion detection in audiovisual narrative. However, several methodological constraints should be acknowledged, along with directions for future research. The analysis focuses exclusively on Season 1 of *Severance*, comprising nine episodes of a single television series within the psychological thriller genre. This limited corpus size restricts the generalizability of findings to other narrative genres, formats, or cultural contexts. The series' distinctive aesthetic, characterized by emotional restraint, minimal dialogue, and thematic focus on

psychological control, may not represent typical patterns of emotional expression in mainstream television. We focused here on creating an analysis framework that may be used to investigate emotions in audiovisual material, and the limitations of using LLMs for this purpose. Future research will be aimed at testing and extending the proposed framework to multiple series across diverse genres (e.g., comedy, drama, documentary) and cultural contexts to assess whether the observed patterns of multimodal emotion expression generalize beyond this specific narrative style.

In terms of annotation, the gold standards were developed by two trained annotators. While this aligns with standard practice in emotion annotation research (e.g., Aman & Szpakowicz, 2007; Bobicev & Sokolova, 2017) and yielded high inter-annotator agreement ($\kappa > 0.80$ at primary and secondary levels), involving additional annotators could provide a more robust measure of annotation reliability and capture a wider range of interpretive perspectives. The union strategy employed during adjudication maximized recall but may have introduced some noise by retaining all emotions identified by either annotator, potentially inflating the emotional diversity of the final gold standards.

The two language models evaluated in this study (GPT-5 and Claude Sonnet 4.5) operated in text-only mode, receiving dialogue scripts without access to visual or auditory information. Recent multimodal LLMs capable of processing video and audio (e.g., GPT-5 with vision, Gemini) could potentially achieve higher alignment with human multimodal annotations. Additionally, this study evaluated only two LLMs from a rapidly expanding ecosystem; testing additional models (e.g., Grok, LLaMA, Mistral, DeepSeek) would provide a more comprehensive picture of current capabilities in emotion detection.

Despite these limitations, the study demonstrates the value of combining human and computational approaches to emotion analysis in audiovisual narrative. The findings underscore the fundamentally multimodal nature of emotional expression and highlight current gaps in LLM capabilities for context-dependent affect recognition. Future research addressing these limitations will further refine our understanding of how emotions are encoded, performed, and perceived in narrative media.

Conclusions

This study set out to examine how emotions are expressed, perceived, and modeled in the TV series *Severance* through a combination of human and computational annotation. Using Parrott's hierarchical taxonomy of emotions, the analysis drew on four distinct datasets (human text-only, human multimodal, and two text-based LLM outputs) to evaluate the influence of modality, embodiment, and algorithmic interpretation on affective understanding.

The results show that human multimodal annotations provided the most complete and ecologically valid picture of emotion in the series, revealing how much affective meaning depends on non-verbal cues (e.g., prosody, posture, gesture, facial expression) rather than words alone. The text-only human annotations, while consistent and reliable, captured only the overt linguistic signals of emotion. In contrast, both LLMs displayed limited but measurable sensitivity to emotional tone, performing best at the

primary level of classification. Their relatively low agreement with human annotators underscores the current limitations of text-based AI models in recognizing affect that relies on pragmatic or multimodal context.

Comparing text-only and multimodal human data confirmed that emotional interpretation is strongly multimodal in nature. When visual and auditory information is available, the range and granularity of recognized emotions expand dramatically, especially at the secondary and tertiary levels. Emotions such as anxiety, frustration, panic, and fondness emerge primarily through performance rather than dialogue, demonstrating that *Severance*'s affective power resides in its understated physicality (e.g., tone, silence, and gaze) rather than lexical choice.

The temporal analysis revealed a pronounced emotional trajectory across the nine episodes. The series evolves from early emotional restraint toward a crescendo of fear, anger, and sadness, alongside brief moments of warmth and solidarity. This gradual escalation mirrors the characters' awakening to their condition and the disintegration of institutional control, resulting in a culmination of affective intensity by the final episodes.

At the level of individual characterization, the multimodal annotations expose how each protagonist embodies a distinct emotional strategy. Helly externalizes rebellion through fear and rage; Mark internalizes grief but rediscovers empathy; Milchick weaponizes affect as control; Irving transforms discipline into longing; and Cobel's repression collapses into fear. Together, they form a small-scale model of emotional control, thus showing how *Severance* explores the struggle to maintain emotional authenticity in an environment designed to suppress it.

Overall, the study demonstrates that emotion in audiovisual narrative cannot be reduced to text. The multimodal dimension is not an embellishment but the primary carrier of affective meaning. Furthermore, the comparison with LLM outputs highlights how current language models, despite their linguistic fluency, remain limited in their capacity to infer human emotion without embodied or contextual grounding. Future research could extend this work by integrating acoustic and visual data directly into annotation pipelines in order to develop multimodal embeddings capable of capturing the full expressive complexity that human annotators can perceive intuitively.

In sum, the multimodal analysis of *Severance* provides a detailed description of how emotions are structured in the series. More generally, it shows that combining computational and human-centered approaches helps to explain how emotions are represented in narrative worlds. The project connects corpus-based annotation with media analysis, thus showing that the study of emotions involves not only what characters say or feel, but also how emotion is shown, interpreted, and communicated through multimodal expression.

Author Contributions A.M.O. designed the methodology, analysed the data, and prepared the figures and tables. M.G.G. compiled the corpus, acted as one of the corpus annotators, and wrote the main manuscript text. All authors reviewed the manuscript.

Data Availability The dataset used in this article is freely available and can be found at https://osf.io/5pbm-c/overview?view_only=14466cb629ae4295aaf2d76b3fb86f41.

Declarations

Conflict of interest The authors declare that there are no competing interests to disclose.

Ethical Approval This is an observational study and it does not involve humans and/or animals.

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References

- Acheampong, F. A., Wenyu, C., & Nunoo-Mensah, H. (2020). Text-based emotion detection: Advances, challenges, and opportunities. *Engineering Reports*, 2(7), e12189. <https://doi.org/10.1002/eng2.12189>
- Al-Azani, S., & El-Alfy, E. S. M. (2025). A review and critical analysis of multimodal datasets for emotional AI. *Artificial Intelligence Review*, 58(10), 334. <https://doi.org/10.1007/s10462-025-11271-1>
- Aman, S., & Szpakowicz, S. (2007). Identifying Expressions of Emotion in Text. In V. Matoušek, & P. Mautner (Eds.), *Text, Speech and Dialogue* (Vol. 4629, pp. 196–205). Springer. https://doi.org/10.1007/978-3-540-74628-7_27.
- An, J., Huang, D., Lin, C., & Tai, M. (2025). Measuring gender and Racial biases in large Language models: Intersectional evidence from automated resume evaluation. *PNAS Nexus*, 4(3), pgaf089. <https://doi.org/10.1093/pnasnexus/pgaf089>
- Anthropic. (2025). *Claude* [Large language model]. <https://claude.ai/login?returnTo=%2F%3F>
- Aue, A., & Gamon, M. (2005). Customizing sentiment classifiers to new domains: A case study. *Proceedings of Recent Advances in Natural Language Processing (RANLP)*, 1, 2–1.
- Bai, X., Wang, A., Sucholutsky, I., & Griffiths, T. L. (2025). Explicitly unbiased large Language models still form biased associations. *Proceedings of the National Academy of Sciences of the United States of America*, 122(8), e2416228122. <https://doi.org/10.1073/pnas.2416228122>
- Balahur, A., Hermida, J. M., & Montoyo, A. (2012). Detecting implicit expressions of emotion in text: A comparative analysis. *Decision Support Systems*, 53(4), 742–753. <https://doi.org/10.1016/j.dss.2012.05.024>
- Bobicev, V., & Sokolova, M. (2017). Inter-Annotator agreement in sentiment analysis: Machine learning perspective. In *RANLP 2017—Recent advances in natural language processing meet deep learning* (pp. 97–102). https://doi.org/10.26615/978-954-452-049-6_015
- Butt, M. A., Qayyum, A., Ali, H., Al-Fuqaha, A., & Qadir, J. (2023). Towards secure private and trustworthy human-centric embedded machine learning: An emotion-aware facial recognition case study. *Computers & Security*, 125, 103058. <https://doi.org/10.1016/j.cose.2022.103058>
- Cambria, E., Livingstone, A., & Hussain, A. (2012). The hourglass of emotions. In A. Esposito, A. M. Esposito, A. Vinciarelli, R. Hoffmann, & V. C. Müller (Eds.), *Cognitive behavioural systems* (Vol. 7403, pp. 144–157). Springer. https://doi.org/10.1007/978-3-642-34584-5_11
- Cambria, E., Das, D., Bandyopadhyay, S., & Feraco, A. (2017). Affective computing and sentiment analysis. In E. Cambria, D. Das, S. Bandyopadhyay, & A. Feraco (Eds.), *A practical guide to sentiment analysis* (pp. 1–10). Springer International Publishing. https://doi.org/10.1007/978-3-319-55394-8_1
- Cambria, E., Poria, S., Hazarika, D., & Kwok, K. (2018). SenticNet 5: Discovering conceptual primitives for sentiment analysis by means of context embeddings. In *AAAI conference on artificial intelligence*. <https://www.aaai.org/ocs/index.php/AAAI/AAAI18/paper/view/16839>

- Cano, S., Díaz-Arancibia, J., Arango-López, J., Libreros, J. E., & García, M. (2023). Design path for a social robot for emotional communication for children with autism spectrum disorder (ASD). *Sensors (Basel, Switzerland)*, 23(11), 5291. <https://doi.org/10.3390/s23115291>
- Chandrasekaran, G., Nguyen, T. N., & Hemanth, D. J. (2021). Multimodal sentimental analysis for social media applications: A comprehensive review. *WIREs Data Mining and Knowledge Discovery*, 11(5), 1415. <https://doi.org/10.1002/widm.1415>
- Chatterjee, A., Narahari, K. N., Joshi, M., & Agrawal, P. (2019). SemEval-2019 Task 3: EmoContext contextual emotion detection in text. In J. May, E. Shutova, A. Herbelot, X. Zhu, M. Apidianaki, & S. M. Mohammad (Eds.), *Proceedings of the 13th international workshop on semantic evaluation*, (pp. 39–48). Association for Computational Linguistics. <https://doi.org/10.18653/v1/S19-2005>
- Cheung, V., Maier, M., & Lieder, F. (2025). Large Language models show amplified cognitive biases in moral decision-making. *Proceedings of the National Academy of Sciences*, 122(25), e2412015122. <https://doi.org/10.1073/pnas.2412015122>
- Chiang, W. L., Zheng, L., Sheng, Y., Angelopoulos, A. N., Li, T., Li, D., Zhang, H., Zhu, B., Jordan, M., Gonzalez, J. E., & Stoica, I. (2024). *Chatbot arena: An open platform for evaluating LLMs by human preference* (No. arXiv:2403.04132). arXiv. <https://doi.org/10.48550/arXiv.2403.04132>
- Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, XX(1), 37–46. <https://doi.org/10.1177/001316446002000104>
- Cowen, A. S., & Keltner, D. (2017). Self-report captures 27 distinct categories of emotion bridged by continuous gradients. *Proceedings of the National Academy of Sciences*, 114(38) 7900. <https://doi.org/10.1073/pnas.1702247114>
- Dahl, M., Magesh, V., Suzgun, M., & Ho, D. E. (2024). Large legal fictions: Profiling legal hallucinations in large Language models. *Journal of Legal Analysis*, 16(1), 64–93. <https://doi.org/10.1093/jla/laae003>
- De Clercq, O., Lefever, E., Jacobs, G., Carpels, T., & Hoste, V. (2017). Towards an integrated pipeline for aspect-based sentiment analysis in various domains. In *Proceedings of the 8th workshop on computational approaches to subjectivity sentiment and social media analysis*, (pp. 136–142). <https://doi.org/10.18653/v1/W17-5218>
- Demszky, D., Movshovitz-Attias, D., Ko, J., Cowen, A., Nemade, G., & Ravi, S. (2020). GoEmotions: A dataset of Fine-Grained emotions (Version 2). *ArXiv*. <https://doi.org/10.48550/ARXIV.2005.00547>
- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional Transformers for Language Understanding. *Proceedings of the 2019 conference of the north*, (pp. 4171–4186). <https://doi.org/10.18653/v1/N19-1423>
- Douglas-Cowie, E., Cox, C., Martin, J. C., Devillers, L., Cowie, R., Sneddon, I., McRorie, M., Pelachaud, C., Peters, C., Lowry, O., Batliner, A., & Hönig, F. (2011). The HUMAINE database. In R. Cowie, C. Pelachaud, & P. Petta (Eds.), *Emotion-Oriented systems* (pp. 243–284). Springer. https://doi.org/10.1007/978-3-642-15184-2_14
- Ekman, P. (1992). An argument for basic emotions. *Cognition and Emotion*, 6(3–4), 169–200. <https://doi.org/10.1080/02699939208411068>
- Fanous, A., Goldberg, J., Agarwal, A. A., Lin, J., Zhou, A., Daneshjou, R., & Koyejo, S. (2025). SycEval: Evaluating LLM sycophancy. *ArXiv*. <https://doi.org/10.48550/ArXiv.2502.08177>. No. ArXiv:2502.08177.
- Gallegos, I. O., Rossi, R. A., Barrow, J., Tanjim, M. M., Kim, S., Dernoncourt, F., Yu, T., Zhang, R., & Ahmed, N. K. (2024). Bias and fairness in large language models: A survey. *Computational Linguistics*, 50(3), 1097–1179. https://doi.org/10.1162/coli_a_00524
- Gao, L., Biderman, S., Black, S., Golding, L., Hoppe, T., Foster, C., Phang, J., He, H., Thite, A., Nabeshima, N., Presser, S., & Leahy, C. (2021). The pile: An 800GB dataset of diverse text for Language modeling (Version 1). *ArXiv*. <https://doi.org/10.48550/ARXIV.2101.00027>
- Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., & Steinhardt, J. (2021). *Measuring Massive Multitask Language Understanding* (No. arXiv:2009.03300). arXiv. <https://doi.org/10.48550/arXiv.2009.03300>
- Henshall, W. (2023). What to know about Claude 2, Anthropic's Rival to ChatGPT. *TIME*. <https://time.com/6295523/claude-2-anthropic-chatgpt/>
- Jiang, Q., Gao, Z., & Karniadakis, G. E. (2025). DeepSeek vs. ChatGPT vs. Claude: A comparative study for scientific computing and scientific machine learning tasks. *Theoretical and Applied Mechanics Letters*, 15(3), 100583. <https://doi.org/10.1016/j.taml.2025.100583>

- Keerthika, M., Abilash, K., Vasanth, M. S., Kuralamudhu, M. K., & Abbiroroban, S. (2024). Emotional AI: Computationally intelligent devices for education. In *emotional AI and human-ai interactions in social networking* (pp. 87–99). Elsevier. <https://doi.org/10.1016/B978-0-443-19096-4.00007-9>
- Khare, S. K., Blanes-Vidal, V., Nadimi, E. S., & Acharya, U. R. (2024). Emotion recognition and artificial intelligence: A systematic review (2014–2023) and research recommendations. *Information Fusion*, 102, 102019. <https://doi.org/10.1016/j.inffus.2023.102019>
- Koptyra, B., Ngo, A., Radliński, Ł., & Kocoń, J. (2023). CLARIN-Emo: Training emotion recognition models using human annotation and ChatGPT. In J. Mikiška, De C. Mulatier, M. Paszynski, V. V. Krzhizhanovskaya, J. J. Dongarra, & P. M. A. Sloot (Eds.), *Computational Science—ICCS 2023* (Vol. 14703, pp. 365–379). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-35995-8_26
- Kumar, S., Mondal, I., Akhtar, M. S., & Chakraborty, T. (2022). *Explaining (Sarcastic) utterances to enhance affect understanding in multimodal dialogues* (Version 2). arXiv. <https://doi.org/10.48550/ARXIV.2211.11049>
- Kumar, S., Akhtar, M. S., Cambria, E., & Chakraborty, T. (2024). SemEval 2024—Task 10: Emotion discovery and reasoning its flip in conversation (EDiReF). In A. Kr. Ojha, A. S. Doğruöz, H. Tayyar Madabushi, G. Da San Martino, S. Rosenthal, & A. Rosá (Eds.), *Proceedings of the 18th international workshop on semantic evaluation (SemEval-2024)* (pp. 1933–1946). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.semeval-1.270>
- Latif, S., Usama, M., Malik, M. I., & Schuller, B. W. (2025). Can large Language models aid in annotating speech emotional data? Uncovering new frontiers [Research Frontier]. *IEEE Computational Intelligence Magazine*, 20(1), 66–77. <https://doi.org/10.1109/MCI.2024.3504833>
- Le Jeune, P., & Berenstein, D. (2025, May 7). *Good answers are not necessarily factual answers: An analysis of hallucination in leading LLMs*. HuggingFace.Com. <https://huggingface.co/blog/davidberenstein1957/phare-analysis-of-hallucination-in-leading-llms>
- Liu, B. (2015). *Sentiment analysis: Mining opinions, sentiments, and emotions*. Cambridge University Press. <https://doi.org/10.1017/CBO9781139084789>
- Mantello, P., Ho, M. T., Nguyen, M. H., & Vuong, Q. H. (2023). Machines that feel: Behavioral determinants of attitude towards affect recognition technology—upgrading technology acceptance theory with the mindsponge model. *Humanities and Social Sciences Communications*, 10(1), 430. <https://doi.org/10.1057/s41599-023-01837-1>
- Minaee, S., Mikolov, T., Nikzad, N., Chenaghlu, M., Socher, R., Amatriain, X., & Gao, J. (2024). Large Language models: A survey (Version 3). *ArXiv*. <https://doi.org/10.48550/ARXIV.2402.06196>
- Mohammad, S. M., & Turney, P. D. (2013). *Crowdsourcing a Word-Emotion Association Lexicon*. <https://doi.org/10.48550/ARXIV.1308.6297>
- Moreno-Ortiz, A. (2017). Lingmotif: A User-focused sentiment analysis tool. *Procesamiento Del Lenguaje Natural*, 58, 133–140.
- Moreno-Ortiz, A., Salles-Bernal, S., & Orrequia-Barea, A. (2019). Design and validation of annotation schemas for aspect-based sentiment analysis in the tourism sector. *Information Technology & Tourism*, 21(4), 535–557. <https://doi.org/10.1007/s40558-019-00155-0>
- Nandwani, P., & Verma, R. (2021). A review on sentiment analysis and emotion detection from text. *Social Network Analysis and Mining*, 11(1), 81. <https://doi.org/10.1007/s13278-021-00776-6>
- Öhman, E. (2021a). Emotion annotation: Rethinking emotion categorization. *Digital Humanities in the Nordic and Baltic Countries Publications*, 3(2), 134–144. <https://doi.org/10.5617/dhnbpub.11241>
- Öhman, E. (2021b). The Validity of Lexicon-based Emotion Analysis in Interdisciplinary Research. In *Proceedings of the workshop on natural language processing for digital humanities (NLP4DH)*, (pp. 7–12).
- Omar, M., Sorin, V., Collins, J. D., Reich, D., Freeman, R., Gavin, N., Charney, A., Stump, L., Bragazzi, N. L., Nadkarni, G. N., & Klang, E. (2025). Multi-model assurance analysis showing large Language models are highly vulnerable to adversarial hallucination attacks during clinical decision support. *Communications Medicine*, 5, 330. <https://doi.org/10.1038/s43856-025-01021-3>
- OpenAI (2025). *ChatGPT* [Large language model]. <https://chat.openai.com/chat>
- Parrott, W. G. (Ed.). (2001). *Emotions in social psychology: Key readings in social psychology*. Psychology.
- Phogat, R., Arora, D., Mehra, P. S., Sharma, J., & Chawla, D. (2025). A Comparative Study of Large Language Models: ChatGPT, DeepSeek, Claude and Qwen. In *2025 3rd international conference on device intelligence, computing and communication technologies (DICCT)*, (pp. 609–613). <https://doi.org/10.1109/DICCT64131.2025.10986449>
- Picard, R. W. (1997). *Affective computing*. The MIT Press. <https://doi.org/10.7551/mitpress/1140.001.0001>

- Plutchik, R. (2001). The nature of emotions. *American Scientist*, 89(4), 344–350.
- Poria, S., Cambria, E., Bajpai, R., & Hussain, A. (2017). A review of affective computing: From unimodal analysis to multimodal fusion. *Information Fusion*, 37, 98–125. <https://doi.org/10.1016/j.inffus.2017.02.003>
- Puschmann, C., & Powell, A. (2018). Turning words into consumer preferences: How sentiment analysis is framed in research and the news media. *Social Media + Society*, 4(3), 2056305118797724. <https://doi.org/10.1177/2056305118797724>
- Pusztahelyi, R. (2020). Emotional AI and its challenges in the viewpoint of online marketing. *Curentul Juridic*, 2, 13–31.
- Rrv, A., Tyagi, N., Uddin, M. N., Varshney, N., & Baral, C. (2024). Chaos with Keywords: Exposing Large Language Models Sycophancy to Misleading Keywords and Evaluating Defense Strategies. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the association for computational linguistics: ACL 2024* (pp. 12717–12733). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.findings-acl.755>
- Russell, J. A., & Mehrabian, A. (1977). Evidence for a three-factor theory of emotions. *Journal of Research in Personality*, 11(3), 273–294. [https://doi.org/10.1016/0092-6566\(77\)90037-X](https://doi.org/10.1016/0092-6566(77)90037-X)
- Socher, R., Perelygin, A., Wu, J., Chuang, J., Manning, C. D., Ng, A., & Potts, C. (2013). Recursive deep models for semantic compositionality over a sentiment treebank. *Proceedings of the 2013 conference on empirical methods in natural language processing*, 1631–1642.
- Stillier, B. (Director) (Ed.). (2022). *Severance [Video recording]*. Apple TV.
- Suzgun, M., Scales, N., Schärl, N., Gehrmann, S., Tay, Y., Chung, H. W., Chowdhery, A., Le, Q., Chi, E., Zhou, D., & Wei, J. (2023). Challenging BIG-Bench Tasks and Whether Chain-of-Thought Can Solve Them. In A. Rogers, J. Boyd-Graber, & N. Okazaki (Eds.), *Findings of the association for computational linguistics: ACL 2023* (pp. 13003–13051). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.findings-acl.824>
- Taboada, M., Brooke, J., Tofiloski, M., Voll, K., & Stede, M. (2011). Lexicon-Based methods for sentiment analysis. *Computational Linguistics*, 37(2), 267–307. https://doi.org/10.1162/COLI_a_00049
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M. A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., Rodriguez, A., Joulin, A., Grave, E., & Lample, G. (2023). LLaMA: Open and efficient foundation Language models (Version 1). *ArXiv*. <https://doi.org/10.48550/ARXIV.2302.13971>
- Van Atteveldt, W., Van Der Velden, M. A. C. G., & Boukes, M. (2021). The validity of sentiment analysis: Comparing manual Annotation, Crowd-Coding, dictionary Approaches, and machine learning algorithms. *Communication Methods and Measures*, 15(2), 121–140. <https://doi.org/10.1080/19312458.2020.1869198>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Proceedings of the 31st international conference on neural information processing systems*, (pp. 6000–6010).
- Wake, N., Kanehira, A., Sasabuchi, K., Takamatsu, J., & Ikeuchi, K. (2023). *Bias in emotion recognition with ChatGPT* (Version 2). *arXiv*. <https://doi.org/10.48550/ARXIV.2310.11753>
- Wang, B., & Liu, M. (2015). *Deep learning for aspect-based sentiment analysis*. *Stanford University Report*.
- Wang, X., Jiang, W., & Luo, Z. (2016). Combination of convolutional and recurrent neural network for sentiment analysis of short texts. In *Proceedings of COLING 2016, the 26th international conference on computational linguistics: Technical papers*, (pp. 2428–2437).
- Xu, Z., Jain, S., & Kankanhalli, M. (2025). *Hallucination is inevitable: An innate limitation of large language models* (No. arXiv:2401.11817). *arXiv*. <https://doi.org/10.48550/arXiv.2401.11817>
- Zahiri, S. M., & Choi, J. D. (2018). Emotion detection on TV show transcripts with sequence-based convolutional neural networks. In *The workshops of the thirty-second AAAI conference on artificial intelligence*.
- Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., & Choi, Y. (2019). HellaSwag: Can a Machine Really Finish Your Sentence? In A. Korhonen, D. Traum, & L. Márquez (Eds.), *Proceedings of the 57th annual meeting of the association for computational linguistics* (pp. 4791–4800). Association for Computational Linguistics. <https://doi.org/10.18653/v1/P19-1472>
- Zhao, G., Li, Y., & Xu, Q. (2022). From emotion AI to cognitive AI. *International Journal of Network Dynamics and Intelligence*, 1, 65–72. <https://doi.org/10.53941/ijndi0101006>
- Zhao, W. X., Zhou, K., Li, J., Tang, T., Wang, X., Hou, Y., Min, Y., Zhang, B., Zhang, J., Dong, Z., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., Liu, Z., & Wen, J. R. (2023). *A survey of large language models* (Version 16). *arXiv*. <https://doi.org/10.48550/ARXIV.2303.18223>

- Zheng, L., Chiang, W. L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023). Judging LLM-as-a-judge with MT-bench and Chatbot Arena. In *Proceedings of the 37th international conference on neural information processing systems*, (pp. 46595–46623).
- Zhou, Z., Asghar, M. A., Nazir, D., Siddique, K., Shorfuzzaman, M., & Mehmood, R. M. (2023). An AI-empowered affect recognition model for healthcare and emotional well-being using physiological signals. *Cluster Computing*, 26(2), 1253–1266. <https://doi.org/10.1007/s10586-022-03705-0>

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